Analyzing Student Pathways Through Process Mining for Curriculum-Aligned Risk Assessment

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Abstract

Universities increasingly rely on learning analytics techniques to support students in understanding their academic progress and making informed study decisions. At the same time, curricular analytics research has shown that the structural design of curricula strongly influences student progression. This paper presents a hybrid analytical framework that integrates process mining with graph-based curricular analytics to provide curriculum-aligned risk assessment of student pathways. We model the curriculum as a weighted prerequisite graph, compute structural progress and adherence metrics using Answer Set Programming, and align student course-taking behavior with a process-mining model derived from the curriculum. These structural and behavioral indicators are combined into the *Student Risk Index* (SRI), a composite metric that quantifies academic risk relative to the expected progression path. Using representative cases, we demonstrate that the SRI effectively distinguishes between students with similar workload progress but different adherence patterns. The results highlight the value of combining curricular and process-mining perspectives to support interpretable, curriculum-aligned student dashboards and early-warning systems.

1 Introduction

Supporting students in understanding their academic progress has become a central objective in higher education. Modern student-facing dashboards increasingly rely on data-driven indicators to help learners interpret where they stand relative to their intended study plan. *Learning Analytics* (LA) provides the methodological foundation for such systems, as it focuses on the "measurement, collection, analysis and reporting of data about learners and their contexts, for purposes of understanding and optimizing learning and the environments in which it occurs" [22]. By transforming raw educational data into actionable insights, LA enables the design of personalized feedback, early-warning mechanisms, and progress-monitoring tools.

However, understanding student progress also requires situating individual learning trajectories within the structure of the curriculum itself. This has led to the emergence of *Curricular Analytics*

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(CA), a complementary field that examines the curriculum as a system – typically represented as a directed graph of courses and prerequisites – to quantify structural complexity, identify bottlenecks, and evaluate how students advance through the prescribed program. CA provides metrics that allow institutions to compare a student's actual path with the idealized curriculum model, offering a deeper understanding of progression patterns and potential obstacles.

A third methodological strand relevant to this work is educational process mining, which converts student activity data into event logs and applies process-mining algorithms to uncover behavioral patterns, deviations, and latent structures in academic pathways. This approach has recently been extended to curricular contexts, where process models derived from student histories can reveal typical and atypical progression routes.

This paper integrates insights from these three domains to develop a framework that informs students about their current standing relative to the curriculum. By synthesizing established metrics from the literature to create a composite **S**tudent **R**isk **I**ndex (SRI), we aim to support the design of dashboards that provide transparent, interpretable, and pedagogically meaningful indicators of academic progress.

1.1 Related Work

1.1.1 Learning Analytics

The field of *Learning Analytics* (LA) has expanded rapidly, with numerous studies examining how student data can be transformed into indicators that support learning and decision-making. Systematic reviews by Viberg et al. (2018) and Banihashem et al. (2018) conclude that LA offers valuable new insights and significant potential for improving education [22, 6]. Beyond the general potential identified in recent reviews, several studies propose specific, measurable indicators that can be used to monitor student progress and institutional performance in higher education. Commonly recommended metrics include credit accumulation and completion rates, which serve as strong predictors of academic progression and degree completion (Tinto, 2014) [21]. Course-level indicators such as course completion rates, pass rates, and grade distributions are also widely used to evaluate student performance and identify at-risk learners (Siemens and Long, 2011) [20]. Early warning systems frequently rely on course performance indicators, including assignment completion patterns and interim grades, to detect students who may require support (Macfadyen and Dawson, 2010; Arnold and Pistilli, 2012) [14, 5]. Additional institutional metrics, such as retention rate, persistence rate, and gateway course completion, provide insight into student progression through key academic milestones and are considered leading indicators of long-term success [13].

1.1.2 Curricular Analytics

Curricular Analytics (CA) conceptualizes academic programs as directed graphs, where nodes represent courses and edges represent prerequisite relationships. This graph-based representation enables the computation of structural metrics that quantify curricular complexity and reveal how curriculum design influences student progression and success. Early work by Ochoa (2016) introduced foundational curricular metrics, including node complexity (the number of prerequisites or dependencies associated with a course), path length (the longest prerequisite chain leading to a course), blocking factor (the number of downstream courses dependent on a given course), and centrality measures that identify structurally critical courses within the curriculum graph. These metrics highlight how structural bottlenecks and long prerequisite chains can create barriers to student advancement [16]. Building on this foundation, Heileman et al. (2018) proposed a comprehensive framework for quantifying curricular reforms. Their approach emphasizes metrics such as the

curricular complexity index, which captures the overall structural difficulty of a program; student curricular efficiency, which measures how closely a student follows the optimal progression path; and bottleneck identification, which identifies courses that disproportionately delay student progression. This framework has become a cornerstone of institutional CA initiatives [11]. Further applications of graph-theoretical approaches appear in Rawatlal (2018), who developed progression maps to demonstrate how structural properties of curricula influence student success patterns [17]. Similarly, Hilliger et al. (2020) focused on designing interpretable and actionable indicators for academic advising dashboards, emphasizing the importance of metrics that support student decision-making and highlight structural barriers within degree pathways [12].

1.1.3 Process Mining Approaches in Education

Process mining (PM) offers a set of techniques for extracting process models from event logs, enabling researchers to analyze how students actually progress through learning activities and academic programs. Foundational work by Aalst (2016) defines the three core families of process-mining methods: process discovery, conformance checking, and enhancement [1]. Systematic surveys by Cairns et al. (2015) and Bogarín, Cerezo, and Romero (2017) highlight the growing use of these techniques in education, emphasizing their value for uncovering authentic student pathways, comparing observed behavior with expected models, and enriching process representations with performance or frequency information [10, 8]. A recent contribution that bridges process mining and curricular analytics is the *Backpack Process Model* (BPPM) proposed by Salazar-Fernandez et al. (2021) [19]. BPPM conceptualizes curricular progression as a process and uses students' course-taking histories as event logs to analyze how learners navigate required and failed courses. This approach enables the extraction of several informative metrics, including path frequency (how often specific curricular routes occur), deviation measures (the extent to which a student's trajectory differs from the expected model), transition probabilities between curricular states, and performance-enhanced indicators that incorporate grades or completion times. These metrics capture the temporal and behavioral dimensions of student progression, complementing traditional curricular-structure metrics by revealing how students actually move through their programs over time.

While *the Backpack Process Model* demonstrates that curricular analytics and process mining can be meaningfully combined to analyze student trajectories, most existing approaches still treat structural curriculum properties and behavioral progression patterns as separate analytical domains. Our work extends this line of research by integrating these perspectives into a unified framework that aligns curricular structure with observed student behavior. Specifically, we compute graph-based structural progress and adherence metrics and couple them with process-mining indicators derived from the curriculum model itself. These complementary signals are combined into the *Student Risk Index* (SRI), which provides a curriculum-aligned assessment of academic risk by capturing both how far students have progressed and how closely their pathways follow the intended sequence.

2 Methodology

The methodology of this study is structured around integrating complementary analytical perspectives to characterize student progression within a curriculum. First, we compute graph-based structural indicators that quantify eligibility, curricular position, and adherence relative to the prerequisite network. Second, we apply process-mining techniques to reconstruct the temporal ordering of students' course-taking behavior and assess their conformance to the curriculum-derived process model. Finally, we combine these structural and behavioral signals into a unified hybrid framework that supports curriculum-aligned risk assessment through the *Student Risk Index* (SRI).

2.1 Student Progress Analysis

Classical measures of Learning Analytics consider the weight of a course based on its credit value. However, in practice, credits alone are not enough to assess a course's importance or its role in blocking student progress. Structural factors, such as prerequisite chains and semester availability, should also be taken into account. For this, we have combined LA with the Curriculum Analytics approach and constructed a curriculum graph from the sample curriculum and formalized prerequisite rules in *Answer Set Programming* (ASP) to compute per-student progress indicators. These indicators are intended to reflect the student's current status as compared to the curriculum.

We represent the curriculum as a directed acyclic graph $G = (V, E)$ where each node $v \in V$ corresponds to a course and each directed edge $(u \to v) \in E$ denotes that course u is a prerequisite of course v. The graph encodes both hard prerequisites (single predecessor required) and composite prerequisites (conjunctive AND sets and disjunctive OR sets).

First, we compute a normalized importance weight $w(v) \in [0, 1]$ for every course v in the curriculum graph so that weights reflect (i) a course's downstream influence, (ii) its position in the curriculum (depth), (iii) its credit value, and (iv) whether it is mandatory. The computation of the weights can be parametrized by α_w, β_w, γ_w, δ_w (summed up to 1) and $\lambda_w \in (0, 1)$ (geometric decay for reachability) parameters which were set empirically to $\alpha_w = 0.35$, $\beta_w = 0.35$, $\gamma_w = 0.2$, $\delta_w = 0.1$, $\lambda_w = 0.8$ to balance structural influence, position, credit value, and mandatory status. The algorithm includes the following steps:

1. For each node v, compute shortest-path distances $d(v \to x)$ to all nodes x reachable from v. The function r() score sums a geometrically decayed contribution from each reachable node:

$$\mathrm{r}(v) = \sum_{x \neq v} \lambda_w^{d(v \to x)-1} \qquad \text{(only if } d \geq 1 \text{ counted).} \tag{1}$$

   This gives a larger value to nodes that reach many successors and weights nearer successors more strongly.

2. For each node v, compute the longest distance to any reachable node:

$$\mathrm{depth}(v) = \max_x d(v \to x) \qquad \text{(0 if no successors.)} \tag{2}$$

3. Normalize the credit value $c(v)$, as well as the mandatory flag $m(v)$ of each course.

4. Define the importance score of each course as a weighted combination of its reach, depth, credit value, and mandatory status.

$$\mathrm{w}(v) = \alpha_w \frac{\mathrm{r}(v)}{R_{\max}} + \beta_w \frac{\mathrm{depth}(v)}{D} + \gamma_w\, c(v) + \delta_w\, m(v), \tag{3}$$

   where $R_{\max}$ and D are used to scale r() and depth() functions into $[0, 1]$. $R_{\max}$ is the maximum of r() scores, and D is the number of semesters in the academic program.

5. Convert $\mathrm{w}(v)$ scores to final normalized weights in $[0, 1]$ by min-max scaling across all nodes.

The r() component captures how many and how close downstream courses depend on v. The depth() component captures how early v, course appears in prerequisite chains, i.e., larger depth means longer chains downstream. High $w(v)$ means that the course is influential, i.e., has many downstream dependents, or deep in the dependency chain, and has high credit, or is mandatory.

The set of course weights is then used to compute per-student progress metrics. To support this computation, *Answer Set Programming* (ASP) is applied to determine, for each student and each semester, the set of available courses based on the prerequisite rules and the student's completed course set . These ASP-derived eligibility sets, together with the graph-based course weights, form the basis of the structural progress indicators described below.

- The total weight of courses completed by a student up to time t:

$$W_{\text{done}}(t) = \sum_{v \in S_t} w(v) \tag{4}$$

  where S_t is the set of courses completed by the student up to semester t.

- The weight of the remaining subjects:

$$W_{\text{rem}} = W_{\text{total}} - W_{\text{done}}(t) \tag{5}$$

  where $W_{\text{total}} = \sum_{v \in V} w(v)$ is the total weight of courses in the academic program.

- The ratio of the remaining weight to the total weight of the program.

$$W_{\text{rem}\%} = \frac{W_{\text{rem}}}{W_{\text{total}}} \tag{6}$$

- Relative adherence to the curriculum that compares the student's completed set S_t to the sample curriculum set M_t expected by the model up to the same elapsed semester:

$$Ra(t) = \frac{|S_t \cap M_t|}{|M_t|}. \tag{7}$$

  $Ra(t) \in [0, 1]$ values close to 1 indicate close adherence to the sample plan. The fraction of program weight that is currently available to the student (eligibility):

$$\mathrm{E}(t) = \frac{\sum_{v \in A_t} w(v)}{\sum_{v \in V} W_{\text{total}}} \tag{8}$$

  where A_t is the set of courses available at time t computed by ASP.

- Effective velocity, i.e., the average completed weight per elapsed semester

$$v_{\text{eff}} = \frac{W_{\text{done}}(t)}{t} \tag{9}$$

  where t is the number of semesters the student has completed.

- Projected finish semester is estimated by combining the remaining curriculum weight with the student's expected future study velocity, which is defined as:

$$v_{\text{exp}} = \mathrm{E}(t)\, v_{\text{eff}} \tag{10}$$

  which adjusts the student's past performance by the structural opportunities available in the upcoming semesters. Using this, the projected finish semester is calculated as:

$$\mathcal{S}_{\text{proj}} = t + \frac{W_{\text{rem}}}{v_{\text{exp}}}. \tag{11}$$

- Projected finish risk, which is a normalized measure of time pressure based on the projected finish semester ($\mathcal{S}_{\text{proj}}$) and program length:

$$\mathcal{R}_{\text{proj}} = \min\left(1,\ \frac{\mathcal{S}_{\text{proj}} - t}{D - t}\right) \tag{12}$$

  For $\mathcal{R}_{\text{proj}} \in [0, 1]$ values near 1 indicate that the projected finish consumes the entire remaining program duration (no slack).

These metrics are complementary: graph/ASP metrics precisely capture eligibility and structural progress, but they do not capture the ordering or concurrency patterns observed in real student progression paths. Therefore, they are most informative when combined with behavioral measures in a hybrid analysis.

2.2 Alignment of student path with curriculum

Temporal factors of student progress can be considered using process mining techniques if student data are transformed into an event log. An event is a record of a single occurrence of an activity related to a case. Formally, an event is a tuple of attribute values. We denote an event by

$$e = \langle \text{case}_{\text{id}},\ \text{act}, \tau, a \rangle \tag{13}$$

where case_{id} is the identifier of the case (student), act is the activity label (enrolled course code), τ is the timestamp, and a is a vector of additional attributes (e.g., semester of enrolment, outcome, and credits). The temporally ordered sequence of events that belong to the same case is called a *trace*:

$$\sigma_c = \langle e_1,\ e_2,\ \ldots,\ e_{n_c} \rangle \qquad \text{with } \ e_i(\text{case}_{\text{id}}) = c \ \text{ and } \ \tau(e_1) \leq \tau(e_2) \leq \ldots \tag{14}$$

The length of a trace n_c is the number of events for case c. The set of traces constitutes an event log: $\mathcal{L} = \{\sigma_c \,|\, c \in C\}$, where C is the set of case identifiers. Equivalently, $\mathcal{L}$ can be viewed as the set of all events partitioned by case_{id}.

In order to align a student trace with the curriculum through process mining methods, the normative model needs to be turned into a process model. Then, conformance checking compares an observed trace (student's sequence of course enrolment events) to a reference process model and quantifies two complementary aspects: fitness (how much the trace can be reproduced by the model) and precision (how much the model allows behavior not seen in the trace). `PM4Py` implements three common techniques for these measures: token-based replay, alignment-based replay, and footprint-based precision; each produces different diagnostics and has different assumptions and costs.

Token replay is fast and scalable and provides direct diagnostic counts, but it is heuristic (inserts tokens) and can over- or under-estimate conformance when the model is not sound [4, 9]. In this method, `PM4Py` simulates the execution of the trace on the *Petri-net* by firing transitions corresponding to trace activities and tracking tokens [18]. When the model cannot fire the required transition, missing tokens are inserted (to continue replay); when the trace ends, remaining tokens may be left in places. A trace fitness is derived from the token bookkeeping: traces that require many inserted tokens or leave many tokens are less fit. The normalized form of fitness is:

$$\mathcal{F}_{\text{trace}} \approx 1 - \frac{\mathcal{T}_{\text{mis}} + \mathcal{T}_{\text{rem}}}{\mathcal{T}_{\text{prod}} + \mathcal{T}_{\text{cons}} + \mathcal{T}_{\text{mis}} + \mathcal{T}_{\text{rem}}} \tag{15}$$

All produced tokens are denoted by $\mathcal{T}_{\text{prod}}$, and tokens that are consumed are denoted by $\mathcal{T}_{\text{cons}}$. Variables $\mathcal{T}_{\text{mis}}$ and $\mathcal{T}_{\text{rem}}$ are the missing and the remaining tokens, respectively. So $F_{\text{trace}} \in [0, 1]$

with 1 means perfect replay. `PM4Py` reports per-trace fitness, percentage of fitting traces, average trace fitness and a log fitness summary.

In alignment-based replay, `PM4PY` computes an optimal sequence of synchronous moves (model+log), model moves, and log moves that transform the model run into the trace. Each deviation has a cost, and the alignment minimizes total cost. The alignment-based fitness is then derived from the optimal cost. This is provably optimal with respect to the chosen cost model and yields exact diagnostics (which model moves correspond to missing or extra events). On the other hand, it is computationally expensive for long traces or large models, and requires a sound model for meaningful alignments [3, 7].

Token-based precision $\mathcal{TP}$ evaluates how restrictive a process model is with respect to the behavior observed in an event log. The metric is based on the escaping edges principle introduced by Muñoz-Gama and Carmona (2010) [15]. Escaping edges represent model behavior that is possible according to the Petri-net but never observed in the log.

Alignment-based precision $\mathcal{AP}$ evaluates how much additional behavior the process model allows beyond what is observed in the event log. Unlike token-based precision, which relies on enabled transitions and escaping edges, alignment-based precision uses optimal alignments between each trace and the model to identify unfit model moves [2].

Footprint precision $\mathcal{FP}$ compares the directly-follows relations (the footprint) observed in the log with those allowed by the model. The value of $\mathcal{FP}$ measures structural over-permissiveness: if the model allows many directly-follows pairs not seen in the log, precision is low. This measure is less costly than alignment-based precision, but it ignores frequency and context of behaviors [9].

The curriculum model used in this study is intentionally highly permissive: it encodes multiple parallel branches (sets of courses that may be taken in arbitrary order or concurrently) and numerous optional branches (elective groups and alternative prerequisite paths). From a modeling perspective, this permissiveness is desirable because it faithfully represents the real flexibility available to students; however, it has important formal consequences for process-centric analyses.

When the curriculum specification is transformed into a *Petri-net*, the resulting net typically satisfies the structural requirements of a well-formed (WF) net. Places and transitions are arranged so that there is a clear start and end region, and the net is syntactically well-constructed. Nevertheless, the same structural permissiveness that models electives and concurrency can make the net not sound in the formal *Petri-net* sense [18]. As a result, alignment-based conformance checking may be infeasible or produce misleading results: the alignment search assumes that every model execution can be completed to the final marking, and when that assumption is violated, the alignment cost minimization becomes ill-posed or computationally unstable. In our experiments, we therefore rely on token-based replay and footprint-based measures as more robust alternatives. These techniques provide useful diagnostics even when the net is permissive and not sound.

2.3 Hybrid approach to quantifying student-to-curriculum conformance

This subsection describes how we combine structural (graph/ASP) progress indicators with process mining (PM) conformance measures to characterize a student's current position relative to the sample curriculum. The combined approach produces an interpretable composite metric that captures both what the student is eligible to do and has done (structural view) and how the student's observed behavior conforms to the modeled curriculum (temporal/behavioral view).

We propose the SRI to quantify a student's academic risk level relative to the expected curriculum progression. It integrates multiple dimensions of study progress, including workload completion, adherence to the recommended curriculum path, predicted time-to-completion, and selected process-mining–based conformance signals.

The SRI is normalized to the interval $[0, 1]$, where: 0 indicates no academic risk (e.g., a student who is on track or has already completed the program), 1 indicates maximum academic risk (e.g., dropout). We define SRI as a convex linear aggregation of structural and behavioral components:

$$\mathrm{SRI} = \alpha\, W_{\mathrm{rem}\%} + \beta\, Ra^{\mathrm{dev}} + \gamma\, \mathcal{R}_{\mathrm{proj}} + \delta\, \mathcal{P}_{\mathrm{on}}^{\mathrm{dev}} + \mu\, \mathcal{F}^{\mathrm{dev}} + \nu\, \mathcal{M} \tag{16}$$

where the weights should be chosen to reflect the relative importance, predictive value, and stability of each component in characterizing academic risk. This way, they ensure that the index remains interpretable and robust across different student trajectories.

In Eq. (6), $W_{\mathrm{rem}\%}$ is the remaining workload ratio that measures how much of the curriculum is still incomplete. $Ra^{\mathrm{dev}}(t)$ is the curriculum adherence deviation, which measures deviation from the recommended curriculum path and is defined as the complement of the relative adherence measure (7) $Ra^{\mathrm{dev}}(t) = 1 - Ra(t)$. $\mathcal{R}_{\mathrm{proj}}$ (12) is the projected finish risk and $\mathcal{P}_{\mathrm{on}}^{\mathrm{dev}}$ is the on-time completion risk, which is the complement of the on-time completion heuristic ($\mathcal{P}_{\mathrm{on}}^{\mathrm{dev}} = 1 - \mathcal{P}_{\mathrm{on}}$). $\mathcal{P}_{\mathrm{on}}$ is defined as a monotonic decreasing function of the delay $\Delta = \mathcal{S}_{\mathrm{proj}} - D$, where $\Delta = 0$ indicates on-time completion. When $\Delta = 0$, the heuristic yields values close to 1, reflecting a high likelihood of on-time completion. As the projected finishing semester moves further beyond the nominal duration, $\mathcal{P}_{\mathrm{on}}$ decreases accordingly. The corresponding risk component is defined as the complement of the heuristic; higher values of $\mathcal{P}_{\mathrm{on}}^{\mathrm{dev}}$ indicate a higher risk of not completing on time. Process mining components are included only as weak signals because they primarily reflect model structure rather than student behavior. Fitness deviation only penalizes very low fitness values: $\mathcal{F}_{\mathrm{dev}} = \max(0; 0.5 - \mathcal{F}_{\mathrm{trace}})$ where $\mathcal{F}_{\mathrm{trace}}$ is defined in (15). Token-based conformance signal denotes the normalized sum of missing and remaining tokens, converted to a binary value:

$$\mathcal{M} = \begin{cases} 0 & M = 0 \\ 1 & M > 0 \text{ (after normalization).} \end{cases} \tag{17}$$

The component weights were established empirically through practical experimentation. After evaluating multiple configurations on real student trajectories, we selected the weighting scheme that provided the most stable, interpretable, and discriminative risk assessment. Workload and adherence receive the highest weights ($\alpha = 0.35$, $\beta = 0.25$) because they are the most stable and meaningful indicators of academic progress. Projected and heuristic completion risks are weighted moderately ($\gamma = 0.1$, $\delta = 0.2$), reflecting their predictive but uncertain nature. Conformance metrics are included only as weak signals ($\mu = 0.07$, $\nu = 0.03$), acknowledging their limited ability to characterize individual student performance.

SRI value is best around $0.0 - 0.2$, meaning that the student is on track with low risk of failing. SRI $\approx 0.2 - 0.5$ indicates moderate risk, some delay or deviation in academic progress. SRI $\approx 0.5 - 0.8$ signals high risk because of significant delays. SRI $\approx 0.8 - 1.0$ predicts very high risk or dropout. The SRI thus provides a unified, interpretable measure of academic progression and risk.

3 Results

The curriculum used in this study corresponds to a seven-semester BSc program in Computer Engineering. It comprises 67 courses in total, including 45 compulsory, 1 optional, and 21 elective courses. As part of the student progress analysis, a directed graph was constructed based on the official curriculum structure (see Figure 1). This graph represents prerequisite relations and the recommended semester-by-semester progression.

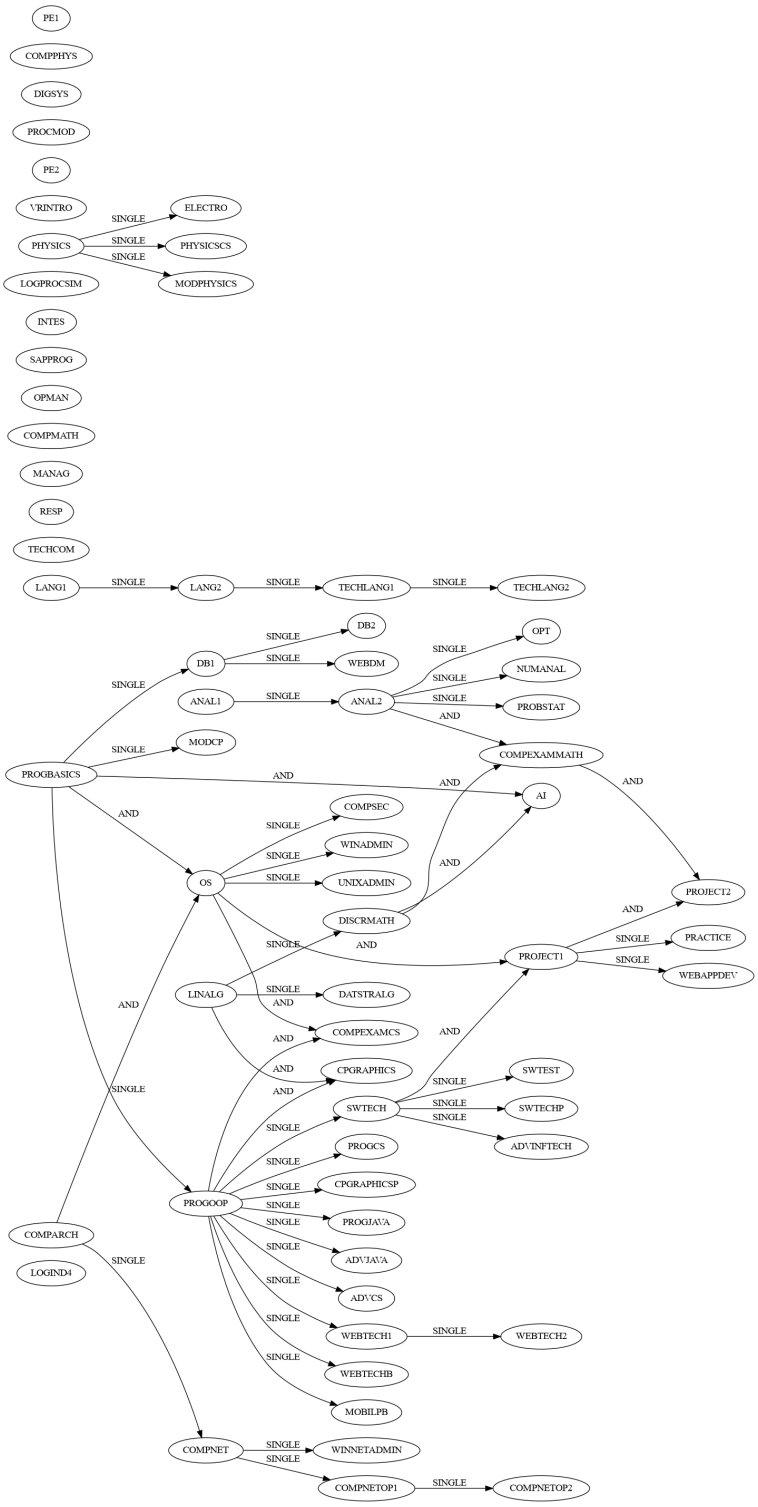

Figure 1: Directed graph of the curriculum structure.

Using this graph, we computed the curriculum weights of all courses as described in Section 2.1, reflecting their structural importance within the program. These weights quantify how strongly each course contributes to overall study progression, based on its position in the prerequisite network

and its role in enabling subsequent courses. After weighing, the most important courses are listed in Table 1 with the recommended semester, the number of prerequisite courses (in-degree), and the number of successor courses (out-degree).

Table 1: List of subjects with the highest weights.

Name	**Type**	**Sem.**	**In_degree**	**Out_degree**	**Weight**
Basics of Programming	Compulsory	1	0	5	1.0
Computer Architectures	Compulsory	1	0	2	0.7201
Lin. Alg. and Discrete Math.	Compulsory	1	0	3	0.6420
Analysis I.	Compulsory	1	0	1	0.6334
Object Oriented Prog.	Compulsory	2	1	11	0.8665
Operating Systems	Compulsory	2	2	5	0.5888
Analysis II.	Compulsory	2	1	4	0.5325
Software Technology	Compulsory	3	1	4	0.5672

Based on the prerequisite graph and the graph-derived course weights, we computed the set of weighted progress indicators defined in 2.1 that quantify each student's advancement within the curriculum structure. These indicators rely directly on the structural importance of courses, as captured by their normalized weights, and form the core quantitative inputs of the Student Risk Index (SRI). These metrics are calculated for five representative students and shown in Table 2. Student 1 successfully completed the entire program in seven semesters (210 credits); Student 2 completed four semesters strictly following the recommended curriculum (120 credits); Student 3 completed four semesters but deviated from the recommended course-taking order (210 credits); Student 4 is not expected to complete the program within seven semesters because it collected only 70 credits in 4 semesters; and Student 5 has dropped out after 4 semesters (60 credits).

Table 2: Weighted progress indicators.

Student ID	Ra^{dev}	W_{total}	W_{done}	W_{rem}	$D-t$	$\frac{W_{\text{rem}}}{D-t}$	$\mathcal{P}_{\text{on}}$	$\mathcal{S}_{\text{proj}}$
1	1.00	17.25	17.25	0.00	0	0.00	1.00	7
2	1.00	17.25	12.12	5.13	3	1.71	0.95	7
3	0.91	17.25	11.92	5.33	3	1.78	0.95	7
4	0.71	17.25	8.65	8.60	3	2.87	0.72	8
5	0.60	17.25	7.22	10.04	3	3.35	0.51	10

The next step was to align each student's academic trajectory with the curriculum model. To achieve this, we constructed an event log from the student records, where each case corresponds to a unique student (`case_id`), and each event represents a course enrolment together with its associated attributes (e.g., semester of course enrolment, relative semester count (`rs`), credits (`cr`), completion status (`cs`), start date of student status, program length (`D`), completion status). The activity label encodes the course code and the number of times the course was taken. Table 3 shows a sample of the resulting event log.

Table 3: Student Log.

case_id	Activity	timestamp	course	enrol_sem.	rs	cr	cs	start_date	D
1	ANAL1_1	2022-09-05	ANAL1	2022/2023/1	1	6	1	2022-09-01	7
1	COMPARCH_1	2022-09-05	COMPARCH	2022/2023/1	1	5	1	2022-09-01	7
1	COMPMATH_1	2022-09-05	COMPMATH	2022/2023/1	1	0	1	2022-09-01	7
1	COMPPHYS_1	2022-09-05	COMPPHYS	2022/2023/1	1	0	1	2022-09-01	7
1	LINALG_1	2022-09-05	LINALG	2022/2023/1	1	6	1	2022-09-01	7
1	OPMAN_1	2022-09-05	OPMAN	2022/2023/1	1	3	1	2022-09-01	7

(rs=relativ_semester, cr=credits, cs=completion_status, D=program_length)

From the curriculum, we created a reference model in the form of a *Petri-net* (see Figure 2). Since within each semester the courses have no prescribed internal ordering or prerequisite relations, the model groups courses by semester, and the Petri-net captures the recommended progression between semesters. As a consequence, the resulting model is a *workflow-net* (WF-net), but it is not easy to understand. Therefore, alignment-based conformance checking cannot be applied.

Instead, we used the token-based replay conformance metrics implemented in the `PM4Py` library. For each student, we computed the token-based fitness and precision values, which quantify how closely the student's actual course-taking behavior follows the recommended curriculum flow. Table 4 summarizes the conformance results for the same five representative students.

Table 4: Conformance results.

Student ID	$\mathcal{F}_{\text{trace}}$	$\mathcal{T}_{\text{miss}}$	$\mathcal{T}_{\text{cons}}$	$\mathcal{T}_{\text{rem}}$	$\mathcal{T}_{\text{prod}}$	$\mathcal{TP}$	$\mathcal{FP}$
1	0.73	52	147	23	0	0.03	0.0065
2	0.73	30	82	11	0	0.03	0.0065
3	0.74	28	94	19	0	0.03	0.0065
4	0.74	21	65	11	0	0.03	0.0065
5	0.74	21	65	11	0	0.03	0.0065

We should note here that precision measures how narrow the reference model is compared to the log. Low values indicate that the model allows a large amount of behavior that does not appear in the log. Curriculum models usually yield low precision because the model permits many possible orderings that students do not actually follow. This way, precision is a property of the model, not of the actual cases. Therefore, in the present experiment, where all students follow the same curriculum, precision does not add any information to student status.

At the same time, fitness measures how well the events in the log can be replayed in the model without missing or remaining tokens. Thus, the fitness value reflects the structural alignment between the model and the log, not the student's academic progress. Shorter traces deviate in fewer places because elective courses have not yet appeared, which is why students who are at the beginning of the program may show slightly higher fitness values.

Produced tokens are zero for all students because the process model is an exact representation of the curriculum, including all prerequisite and dependency constraints. These rules are strict and cannot be violated during replay.

As the last step, we compute the SRI and its components for the sample students. These results are collected in Table 5.

Table 5: SRI and its components.

Student ID	SRI	`status`	$W_{\text{rem}\%}$	Ra^{dev}	$\mathcal{R}_{\text{proj}}$	$\mathcal{P}_{\text{on}}^{\text{dev}}$	$\mathcal{F}^{\text{dev}}$	$\mathcal{M}$
1	0.00	finished	0.00	0.00	0.00	0.00	0.0	0.00
2	0.13	active	0.30	0.00	0.00	0.05	0.0	0.63
3	0.16	active	0.31	0.09	0.00	0.05	0.0	0.70
4	0.34	active	0.50	0.29	0.14	0.28	0.0	0.68
5	1.00	dropout	1.00	1.00	1.00	1.00	0.0	0.00

4 Discussion

In this section, we illustrate how the Student Risk Index (SRI) provides a more informative and discriminative assessment of student progress than any single progress- or conformance-based indicator. To demonstrate this, we selected a small set of representative students from the full dataset

of 610 cases, each exhibiting distinct patterns of advancement and behavioral alignment with the curriculum model. By comparing these students across multiple dimensions (progress, adherence, structural conformance, and behavioral deviations), we show that the SRI integrates these aspects into a coherent measure that clearly separates students who appear similar when viewed through individual indicators. The following figures highlight how the SRI captures meaningful differences in student trajectories that remain hidden when relying solely on traditional progress metrics or token-based conformance measures.

4.1 Student progress

Figure 3 compares students based on how much of the curriculum they have completed (W_{done}) and how much workload remains relative to the full program ($W_{\text{rem}\%}$). Students 2 and 3 have both completed four semesters and appear close to each other in this representation. Although their overall progress is similar, this plot does not reveal whether they follow the recommended course sequence or deviate from it.

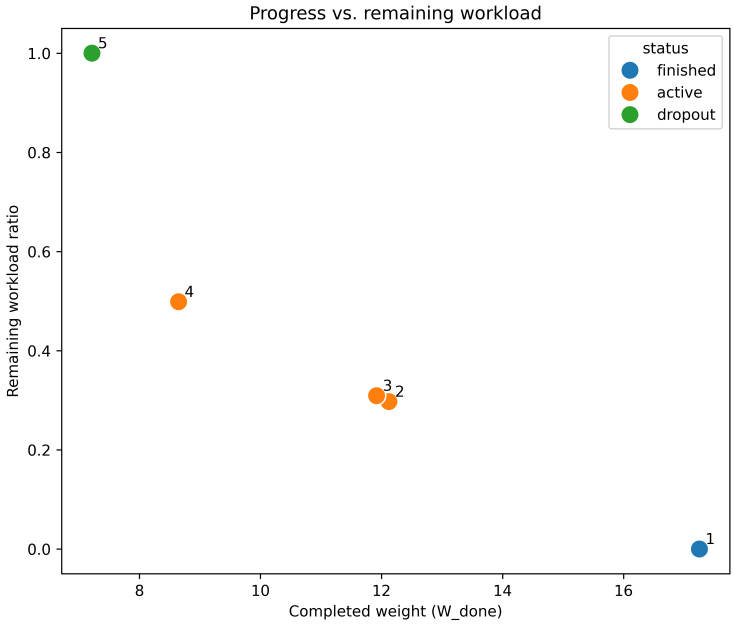

Figure 3: Progress vs. Remaining Workload.

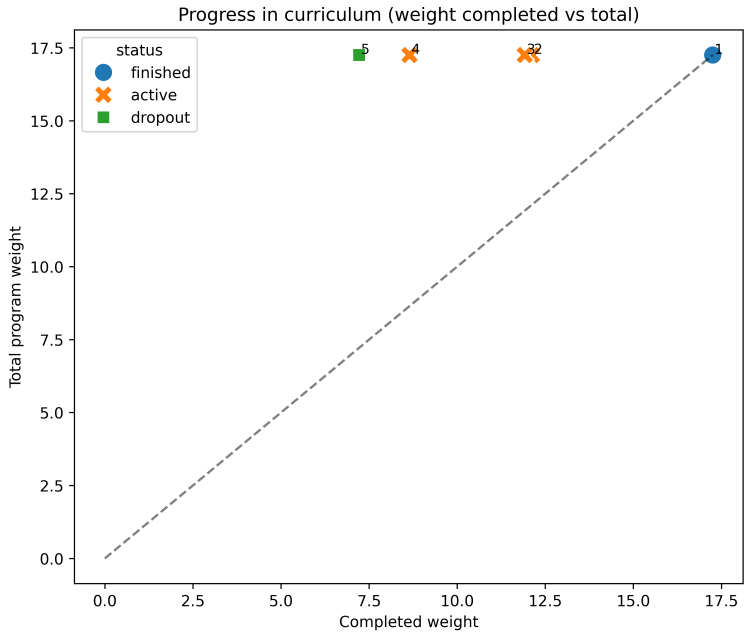

Figure 4: Structural Progress.

Structural progress measures how far a student has advanced in the curriculum structure, independent of the actual course order taken (see Figure 4). Students 2 and 3 again appear close, as both have completed the same number of semesters. However, this metric still does not capture whether their course-taking behavior aligns with the curriculum structure.

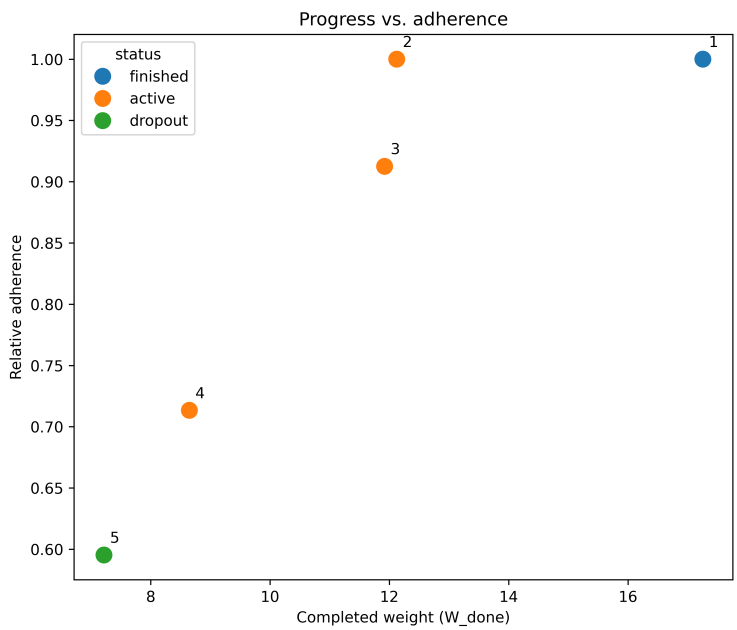

Figure 5: Progress vs. Adherence.

Figure 5 contrasts curriculum progress (W_{done}) with relative adherence (Ra), which quantifies how closely a student follows the recommended course-taking order. Here, Students 2 and 3 become

clearly distinguishable: Student 2 follows the curriculum sequence closely, while Student 3 deviates from it despite having completed a similar amount of coursework. This plot demonstrates that adherence is a more sensitive indicator of structural deviations than progress alone.

4.2 Student conformance

Figure 6 compares students based on token anomalies generated during replay. Missing tokens indicate deviations where the log requires behavior not enabled by the model, while remaining tokens indicate unused model behavior. Students 4 and 5 overlap in this representation, because both produce similar amounts of token anomalies. However, their academic situations differ substantially: Student 4 is still active but progressing far from the curriculum structure, whereas Student 5 has already dropped out. Token anomalies alone cannot distinguish these cases.

Figure 7 separates structural risk (deviation from the curriculum structure) from behavioral risk (deviation from the model's allowed behavior). Unlike the token anomaly plot, this representation clearly distinguishes Students 4 and 5. Student 4 exhibits high structural risk due to taking courses in an order inconsistent with the curriculum, while Student 5 shows high behavioral risk associated with dropout behavior. This plot demonstrates that structural and behavioral risk capture different dimensions of conformance.

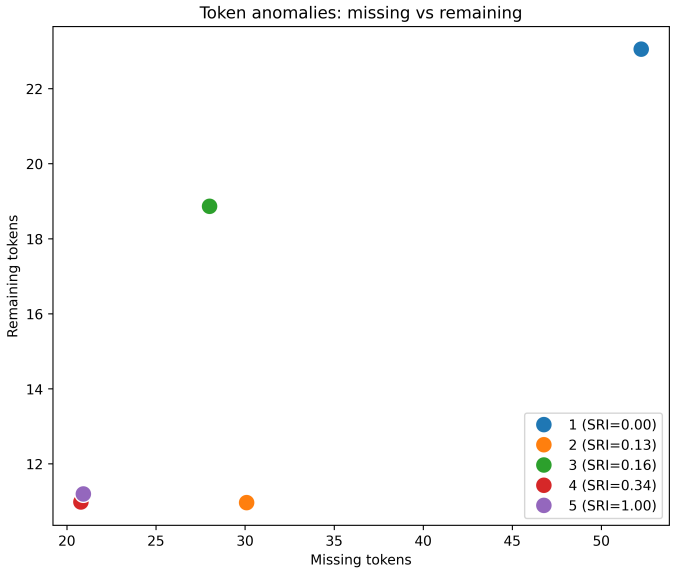

Figure 6: Token Anomalies ($\mathcal{T}_{\text{miss}}$ Missing vs. $\mathcal{T}_{rem}$ Remaining Tokens).

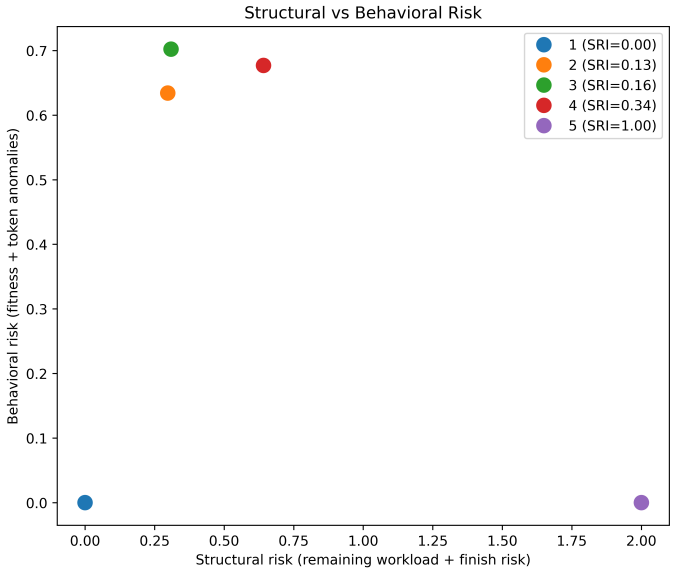

Figure 7: Structural vs. Behavioral Risk.

4.3 Combined metrics (SRI)

The Student Risk Index (SRI) aggregates progress, adherence, and conformance indicators into a single risk score (see Figure 8). The ranking highlights which students are at higher risk relative to the curriculum model. Students with high SRI values either deviate substantially from the curriculum structure or exhibit behavioral patterns associated with dropout.

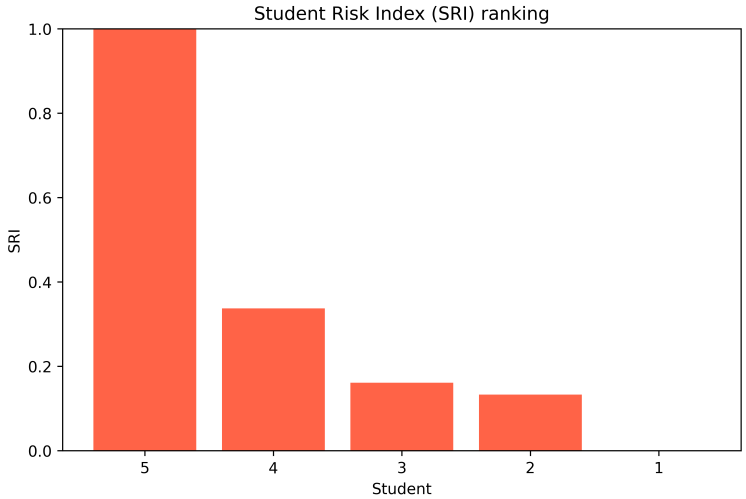

Figure 8: SRI Ranking.

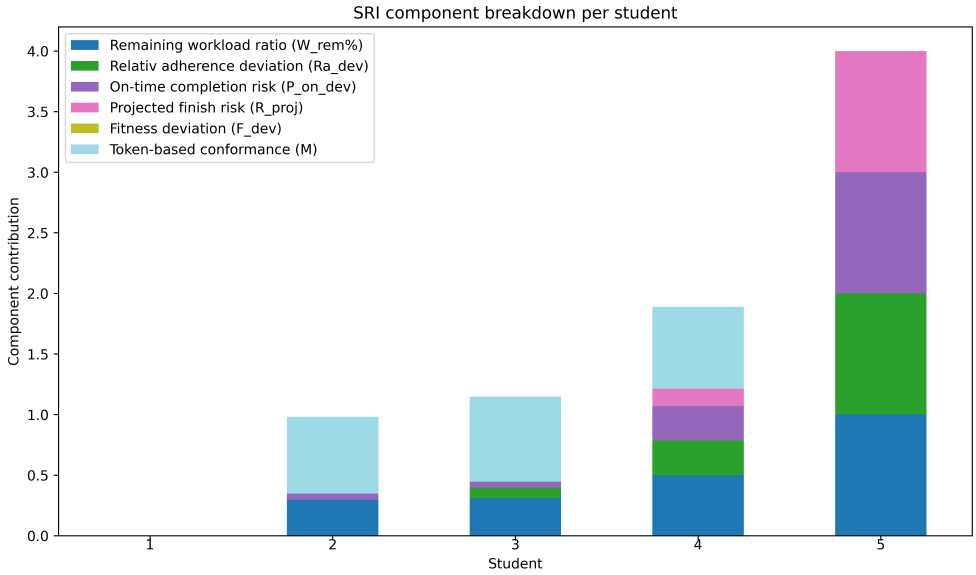

Figure 9: SRI Components.

Figure 9 decomposes the SRI into its structural, behavioral, and progress-related components. The visualization shows how different factors contribute to each student's overall risk level. Students with similar progress may still differ significantly in structural or behavioral risk, illustrating why a combined metric is necessary to capture the full picture of student performance relative to the curriculum.

The high SRI values observed for some students are primarily driven by elevated levels of $W_{\text{rem}\%}$, Ra^{dev}, $\mathcal{R}_{\text{proj}}$, and $\mathcal{P}_{\text{on}}^{\text{dev}}$. As these component metrics decrease, the overall SRI score also decreases, reflecting lower structural and behavioral risk relative to the curriculum model. Among active students, the $\mathcal{M}$ component also appears in the SRI decomposition, and its contribution is relatively similar across the examined cases.

This consistency in $\mathcal{M}$ arises because active students typically generate comparable amounts of model-driven behavior during replay. Since they are still progressing through the curriculum, their traces contain a similar proportion of model-enabled transitions, which results in a normalized model-alignment component of roughly equal magnitude. In other words, $\mathcal{M}$ does not differentiate strongly between active students at this stage, because their interaction with the model structure is still relatively uniform.

4.4 Summary

Together, these figures demonstrate that the SRI provides a comprehensive view of a student's position relative to the curriculum. While progress indicators alone cannot distinguish between students who follow or deviate from the recommended sequence, and conformance indicators alone cannot separate structural deviation from dropout behavior, the combined SRI metric integrates these dimensions into a coherent risk assessment.

5 Conclusions

This study demonstrated how combining *Learning Analytics*, *Curricular Analytics*, and *Educational Process Mining* provides a multidimensional understanding of student progression within a struc-

tured academic program. By integrating graph-based curriculum weights, ASP-derived prerequisite constraints, and process-mining conformance signals, we developed the Student Risk Index (SRI), a composite metric that captures both structural and behavioral aspects of academic advancement. Using five representative students selected from a dataset of 610 cases, we showed that traditional progress indicators are insufficient to distinguish between students who advance at a similar pace but follow different course-taking patterns. Likewise, token-based conformance metrics alone cannot differentiate between structural deviation and dropout behavior.

The SRI resolves these limitations by synthesizing workload completion, adherence to the recommended curriculum, projected time-to-completion, and weak conformance signals into a single interpretable score. The visual analyses presented in Section 4 of this work illustrate how the SRI clearly separates students with superficially similar progress but fundamentally different academic trajectories. As such, the SRI provides a more discriminative and actionable measure of academic risk, supporting the design of student-facing dashboards and early-warning systems.

6 Limitations

Although the SRI framework offers a comprehensive view of student progression, several limitations should be acknowledged. First, the curriculum model used in this study is based on a single academic program with a well-defined prerequisite structure. Programs with more flexible or less formally specified curricula may require additional modeling assumptions or alternative representations. Second, the process model derived from the curriculum is intentionally permissive, which limits the computation and interpretability of conformance metrics.

Third, the SRI relies on heuristic estimates of on-time completion and projected finish semester, which may be sensitive to local institutional policies, course availability patterns, or student-specific factors not captured in the event log. Fourth, the evaluation was conducted on a limited set of representative students; although these cases were selected from a larger dataset, a broader empirical validation is needed to assess the generalizability and predictive value of the SRI across diverse student populations.

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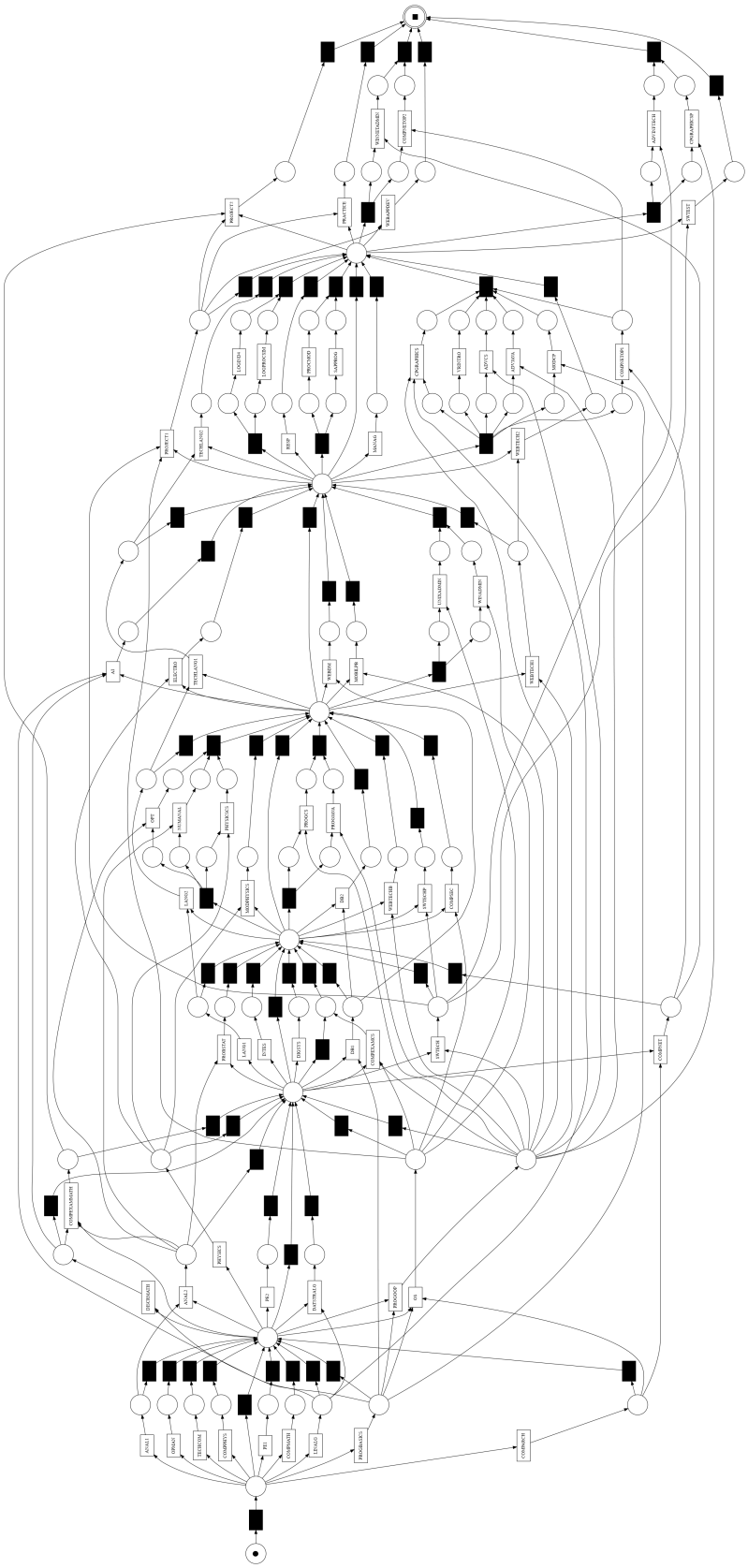

Figure 2: Curriculum structure in the form of Petri-net.